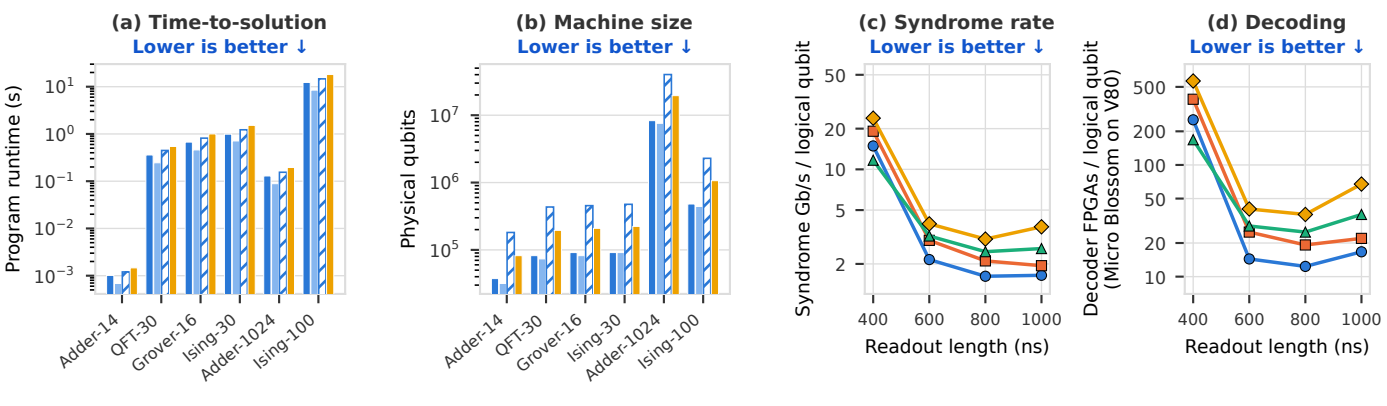
(a) Time-to-solution
Lower is better ↓
Program runtime (s)
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
Adder-14
QFT-30
Grover-16
Ising-30
Adder-1024
Ising-100
(b) Machine size
Lower is better ↓
Physical qubits
10^{7}
10^{6}
10^{5}
Adder-14
QFT-30
Grover-16
Ising-30
Adder-1024
Ising-100
(c) Syndrome rate
Lower is better ↓
Syndrome Gb/s / logical qubit
50
20
10
5
2
400
600
800
1000
Readout length (ns)
(d) Decoding
Lower is better ↓
Decoder FPGAs / logical qubit
(Micro Blossom on V80)
500
200
100
50
20
10
400
600
800
1000
Readout length (ns)
CNN, 1000 ns
CNN, 600 ns
CNN, 400 ns
QubiCML, 1000 ns
MCMit-CNN
HERQULES
KLiNQ
QubiCML